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Re: NSF Pathways to Enable Secure Open-Source Ecosystems (PESOSE) Program Application

To Whom It May Concern:

I am writing on behalf of Dr. Jonathan Matthis, with whom I have had an ongoing collaboration with through the FreeMoCap Foundation since 2024.

Our lab studies the development of natural visual and sensorimotor behavior in the freely moving ferret. Specifically, we are focused on the rapid development of oculomotor skills after eye opening. Our recording system and analysis pipeline **required simultaneous measurements of natural behavior and neural activity**, including binocular eye movements, three-dimensional head and body movement, and recordings from visual cortex. Because animals are developing, markerless motion-capture is the only available method and standard human workflows would not work. Thus, in a collaboration with the FreeMoCap team, a 3D reconstruction and analysis pipeline using multiple camera systems was built. **Without FreeMoCap our work would not be possible.**

One component of this pipeline is SkellyClicker, an open-source tool that makes custom DeepLabCut tracking models easier to create and use within motion-capture workflows. The FreeMoCap team led its development, with individual member of my lab contributing to design, testing, and day-to-day usability. Because SkellyClicker supports custom tracking models, the same workflow can be adapted for other researchers working with nonhuman animals or other tracking problems that are not well served by standard human pose-estimation models.

With funding from PESOSE project, our collaboration with the FreeMoCap ecosystem can accelerate— as both scientific users and contributors. Our lab expects to continue contributing to the development and evaluation of tools and workflows for nonhuman tracking, while providing feedback on technical interfaces, documentation, and contributor processes that can make specialized extensions easier to develop and share. We are excited to continue this collaboration and know that through it, we are able to tackle leading scientific questions and develop novel unique research directions.

Sincerely,

Benjamin Scholl, Ph.D.

Assistant Professor

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